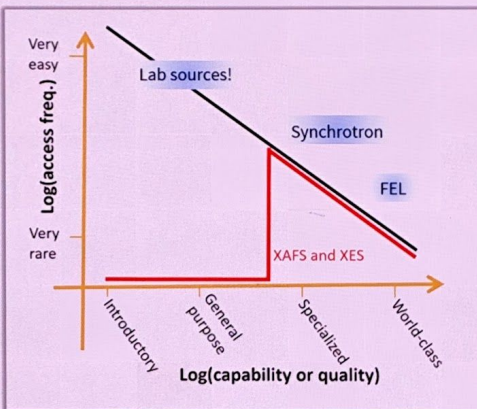
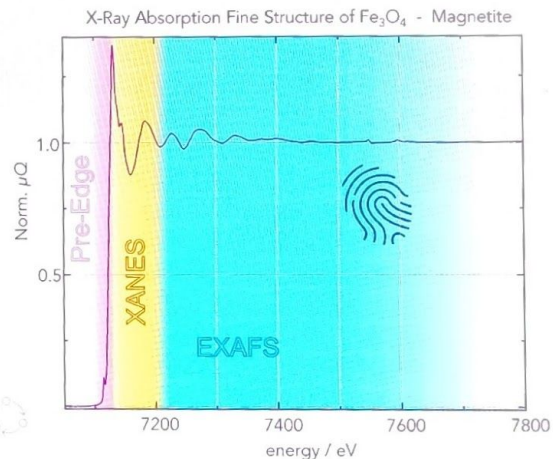


XAS in Laboratorio

Motivation

Why *operando* and *in situ* XAS in laboratory?

- Real-time monitoring of **local electronic structure**, **oxidation states**, and **coordination environments** during reactions/cycling
→ investigation of **phase transition**, **redox mechanism**,...
- Important in
 - Catalyst screening & optimization
 - Battery degradation mechanism
 - ...
- Advantage of laboratory setups
 - Flexible "beam" time
 - Quick feedback cycles → rapid material development
 - Sample pre characterisation



G. Seidler, <https://youtu.be/3IJ9uE7Xvcg>

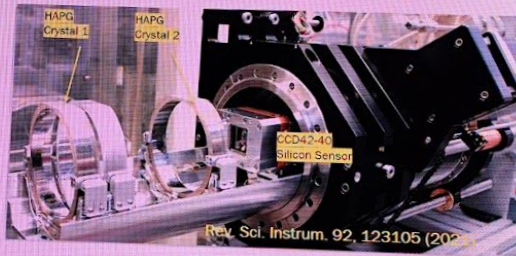
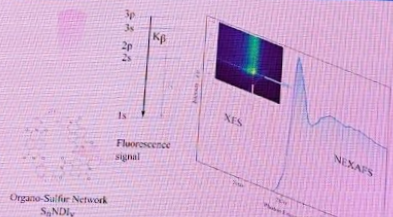
Ottimizzazione per Operando XAS in Laboratorio

X-ray Emission Spectroscopy (XES) with a von Hamos Spectrometer

SPP 2248
Polymer-based
Batteries

UNI
FREIBURG

Freiburg Institute for
Materials Research



XES measurements with Von Hamos Spectrometer

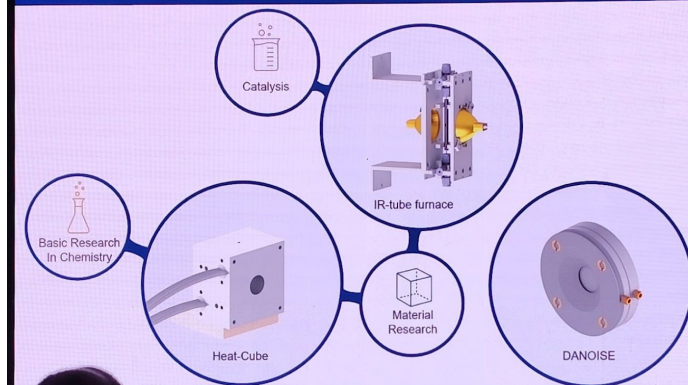
- Full cylinder design → high detection efficiency
- Wavelength-dispersive, scan-free → high reproducibility
- Broad photon energy range from 2.4 keV to 18 keV (1 crystals), 2.3 keV to 13 keV (2 crystal)
- Non-resonance XES are free from self-absorption effect, thus convenient for quantitatively analysis

PTB

EXRS 2026

Hongfei Yang, 15.06.2026 5

The three *in situ/operando* setups

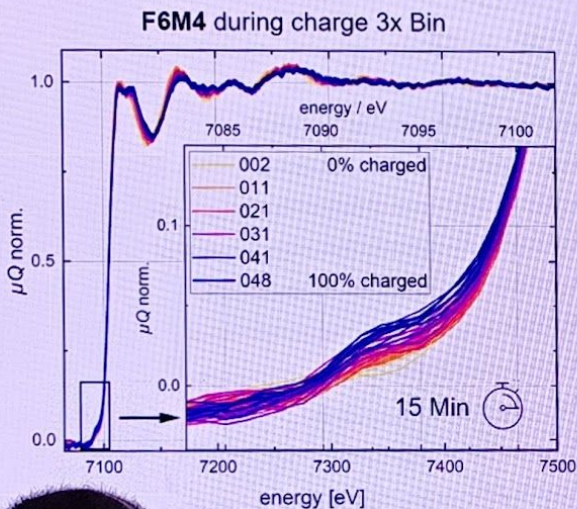
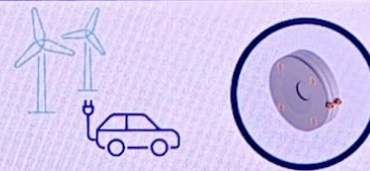


GRANT GIOVANI IN CSN5

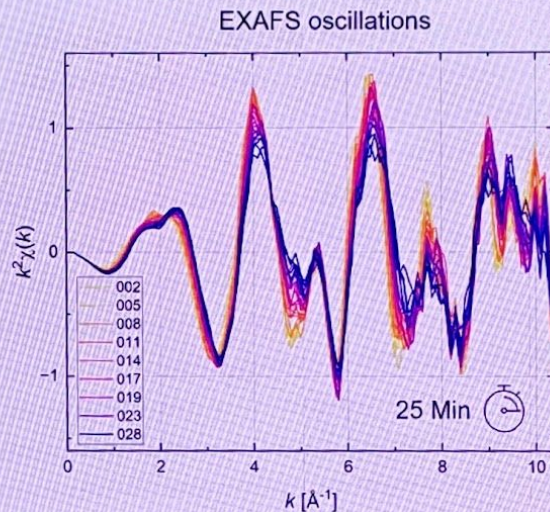
Operando XAS on Na-batteries

Sodium(Na)-ion layered oxides as complex systems at the **Fe** K-edge

- Na-ion batteries offer lower costs, higher abundance, and reduced environmental impact →
- $\text{Na}_{0.72}\text{Fe}_{0.6}\text{Mn}_{0.4}\text{O}_2$ (**F6M4**) measured during **12 h** of charging



5x Bin



XAS in Fluorescenza in Laboratorio

Interior Schematic of QuantumLeap-Series

Patented Advantages

1. Patented High Brightness X-ray Source with Multiple Targets

Multiple X-ray targets:

- W
- Rh
- Cu
- Fe
- Cr

2. Patented System Design to allow for low bragg Angle Operation

3. Dual Detectors (fluorescence and transmission mode XAS)

Silicon Drift Detector (SDD) for fluorescence-mode XAS

Johansson Crystal Analyzer

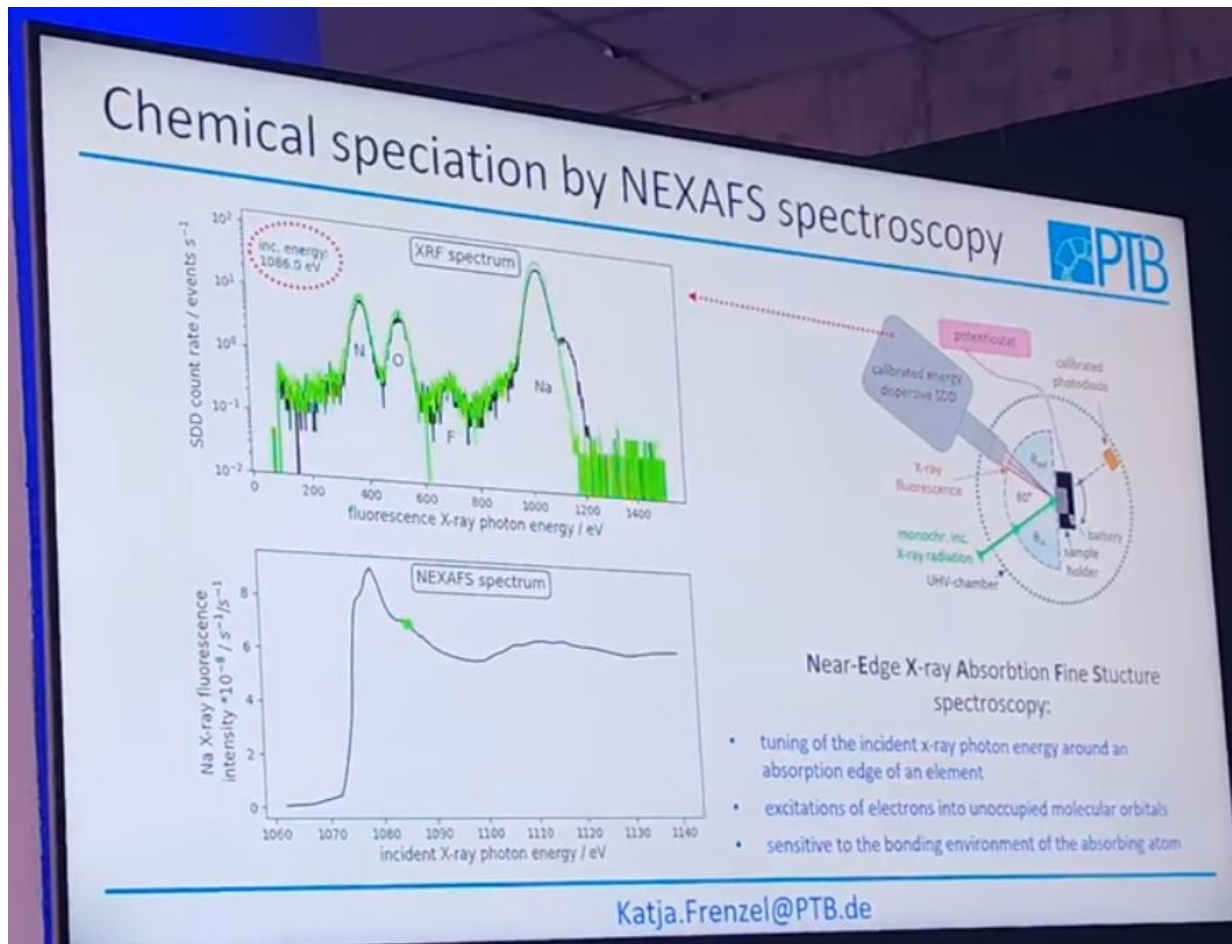
Monochromated Beam

Sample

Spatially Resolving Photon Counting Detector for transmission mode XAS

SIGRAY

XAS in Fluorescenza in Laboratorio



- No preparazione campione o fogli sottili
- Misura da picchi fluorescenza con SDD
- Scan angolare con cristallo per selezione energia

GRANT GIOVANI IN CSN5

- **XAS in laboratorio:**
 - Trasmissione: più facile ma preparazione campione
 - Fluorescenza: no preparazione campione
- **Sorgenti Estese** vs Microfocus
- **XAS da laboratorio**-> Setup Lab-based assorbimento in CSN5
- **Fluorescence Mode**-> Sviluppo SDD
- **Motori Vuoto + Sorgenti Estese**-> EuAPS